Schur Covariance Evaluation

A Principled Pseudo-Likelihood in High Dimensions

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Abstract

We introduce the **Schur pseudo-likelihood**, a one-parameter family ℓ_γ, $\gamma \in [0,1]$, that deforms the Gaussian likelihood by damping the cross-block coupling carried by its Schur complements. The endpoints are familiar — $\gamma=1$ is the exact joint likelihood, $\gamma=0$ is block composite (pseudo-)likelihood — and the interior is the new object: a continuous **coupling-strength** dial that the composite-likelihood, Vecchia, and Gaussian-Markov-random-field literatures do not have, because those methods regularize the *conditioning structure* (which variables to condition on) at full strength, an axis orthogonal to γ. The contribution is deliberately narrow and, we think, clean: not a new estimator but a *pseudo-likelihood with a single interpretable knob* that (i) recovers the power and conditioning the full likelihood loses in high dimensions, where it becomes — we show — worse than chance at ranking estimators; (ii) admits an exact characterization of what γ does, via a worked model we verify numerically: it recovers the true predictive law for every γ but maps to an inflated covariance that leaves the positive-definite cone below a coupling-dependent threshold — pinning down precisely why ℓ_γ is a device for *scoring* or *transforming* a covariance rather than for estimating one by free maximization, and for which the optimal trust has a closed form — $\gamma^* = (n-2)\rho^2/[(n-2)\rho^2+(1-\rho^2)]$, the *reliability* of the coupling (a Wiener/James–Stein shrinkage), rising to 1 as the sample size or coupling grows and to 0 as the coupling becomes noise; and (iii) is **broadly applicable** — wherever a Gaussian block likelihood is used (spatial statistics, graphical models, longitudinal data, any composite-likelihood setting), and even, via the identical algebra, in portfolio allocation, where the same γ interpolates hierarchical risk parity and minimum-variance optimization. We show in simulation that an interior γ strictly dominates both endpoints when the coupling is partially reliable, that a geodesic (affine-invariant) damping is stable where the linear one is not, and that the high-dimensional inversion catastrophe the construction avoids is borne out on real equity returns.

1. The high-dimensional failure of the likelihood

For a zero-mean Gaussian the per-sample log-likelihood is $\ell(\Sigma;\ x) = -½(\log\det\Sigma + x^\top\Sigma^{-1}x)$, equivalently the **covariance-likelihood** form in the sample covariance S, $\ell(\Sigma;\ S) = -½(\log\det\Sigma + \mathrm{tr}(\Sigma^{-1}S))$ — proportional to the Wishart log-likelihood of S given Σ and, up to constants, the negative Stein/entropy loss. Both terms are governed by the *smallest* eigenvalues of Σ: $\log\det = \Sigma \log \lambda_i \to -\infty$ and Σ^{-1} blows up as $\lambda\_min \to 0$. When $p/n \to c > 0$ (the Marčenko–Pastur / spiked regime) those eigenvalues collapse toward zero and are unidentifiable, so the likelihood stakes its verdict on noise. Empirically (`research/metric_power.py`) the held-out likelihood drops to ≈**0.45** — *below chance* — at correctly ordering two estimates of *known* relative quality, while inversion-free judges hold ≈0.85. This is the evaluation-side image of Markowitz "error maximization" in $w \propto \Sigma^{-1}1$: the same Σ^{-1}, the same fragility. The remedy is to stop trusting the raw inverse — the question is how, with one interpretable knob.

2. The exact Schur factorization of the Gaussian likelihood

Partition the p variables into K ordered blocks $x = (x\_1,\ \ldots,\ x\_K)$. The joint density factorizes *exactly* into block conditionals, $p(x) = \prod\_k\ p(x\_k \mid x\_\{<k\})$, and for a Gaussian each is itself Gaussian with

- conditional covariance = **Schur complement** $S\_k = \Sigma\_kk - \Sigma\_\{k,<k\}\ \Sigma\_\{<k,<k\}^{-1}\ \Sigma\_\{<k,k\}$,
- conditional mean $\mu\_\{k|<k\} = \mu\_k + \Sigma\_\{k,<k\}\ \Sigma\_\{<k,<k\}^{-1}\ (x\_\{<k\} - \mu\_\{<k\})$.

Hence $\log\det\Sigma = \Sigma\_k\ \log\det S\_k$ and the quadratic form splits blockwise — the block-LDL$^\top$ / block-Cholesky factorization, the engine of Vecchia, GMRF, and nested-dissection elimination. **No approximation has been made yet**: only a re-expression of the same likelihood as a sum of block-conditional terms, each inverting only the conditioning block $\Sigma\_\{<k,<k\}$.

3. The Schur pseudo-likelihood

We introduce a single **coupling-strength** parameter $\gamma \in [0,1]$ that damps the Schur complement and the conditional regression in lockstep:

```
S_k(γ)      = Σ_kk − γ · Σ_{k,<k} Σ_{<k,<k}⁻¹ Σ_{<k,k}   =  (1−γ) Σ_kk + γ S_k
μ_{k|<k}(γ) = μ_k + γ · Σ_{k,<k} Σ_{<k,<k}⁻¹ (x_{<k} − μ_{<k})

ℓ_γ(Σ; x)   = Σ_k  log N( x_k ; μ_{k|<k}(γ), S_k(γ) ).
```

γ	ℓ_γ is	character
1	the **exact joint Gaussian likelihood**	most powerful, fragile (inverts all of Σ)
$(0,1)$	a **tunable bridge** that *changes the block covariances*	inverts only conditioning blocks; better-conditioned
0	the **block composite likelihood**	robust; product of block marginals

What kind of object is ℓ_γ? At $\gamma=1$ it is the joint log-density; at $\gamma=0$ it is a product of block marginals — a composite likelihood in the sense of Besag/Lindsay/Varin–Reid–Firth: a sum of proper component log-densities that does not itself integrate to one. For $\gamma\in(0,1)$ each term $\log N(x\_k;\ \mu(\gamma),\ S\_k(\gamma))$ is still a *genuine* Gaussian log-density (a strictly proper local score whenever $S\_k(\gamma) \succ 0$), but the damped conditionals are mutually inconsistent — they are not the conditionals of any single joint Gaussian — so ℓ_γ is a **pseudo-likelihood**, not a normalized joint. This is the composite-likelihood bargain: trade the joint-density property for tractable, well-conditioned components, and γ controls how much of the cross-block information the marginals discard is dialed back in.

Three uses, only two of them well-behaved. (a) *Score* a fixed estimate Σ by $\ell\_\gamma(\Sigma;\ S\_test)$. (b) *Transform* a given Σ by replacing each conditional covariance with $S\_k(\gamma) = (1-\gamma)\Sigma\_kk + \gamma\ \hat{S}\_k$ — a structured (block-conditional) **shrinkage** of the coupling toward block-diagonal with intensity $1-\gamma$, the basis-dependent counterpart of spectral shrinkage (§6); always PSD for $\gamma\in[0,1]$ and the operation `SchurCovariance` and Schur allocation perform. (c) *Estimate* Σ by maximizing $\ell\_\gamma(\Sigma;\ S)$ — which §4 shows is **degenerate** for $\gamma<1$. The clean reading is that ℓ_γ is a scoring/transform device, not a free estimation objective.

4. What γ does, exactly: a worked model

To see precisely what damping does — and why estimation by free maximization fails while scoring and transformation do not — take the smallest non-trivial case and solve it in closed form. Let $x_1 \sim N(0,\ a)$, $x_2 = b\star\ x_1 + \epsilon$, $\epsilon \sim N(0,\ s\star)$, two scalar blocks, so the true conditional law of x_2 given x_1 has regression $b\star$ and Schur complement (conditional variance) $s\star$. For a candidate $\Sigma = (A,\ C,\ D)$ the damped conditional has effective coefficient $\beta = \gamma C/A$ and variance $S(\gamma) = D - \gamma C^2/A$. Maximizing the population (expected) ℓ_γ over $(A,\ C,\ D)$ gives (verified numerically in `research/schur_likelihood_theory.py`):

> **Proposition (two-block Gaussian).** For every $\gamma \in (0,1]$ the maximizer recovers the *true predictive law* — $\beta = b\star$ and $S(\gamma) = s\star$ — but the *implied covariance* is **inflated**: $\hat{A} = a$, $\hat{C} = \Sigma_{12}\star\ /\ \gamma$, $D = \Sigma_{22}\star + b\star^2 a\ (1/\gamma - 1)$. This Σ is positive-definite iff $\gamma^2\ (1 - \rho^2) > (1 - \gamma)\ \rho^2$, where $\rho^2 = b\star^2 a\ /\ (b\star^2 a + s\star)$ is the block coupling (the conditional R^2). Below the threshold $\gamma\_min(\rho^2)$ the free maximizer leaves the PSD cone.

The numerics ($a=1,\ b\star=0.8,\ s\star=0.5$, so $\rho^2=0.561$) show $\beta=0.800$ and $S(\gamma)=0.500$ for every γ while $\hat{C}$ runs $0.80 \to 8.0$ as $\gamma: 1\to 0.1$, and the maximizer is PSD only above $\gamma\_min = 0.660$.

The general case (verified in `research/schur_likelihood_theory.py` by an exact closed-form expected ℓ_γ and independent local search):

> **Proposition (general blocks).** For an arbitrary ordered partition the population ℓ_γ maximizer recovers *every block's true conditional law* — effective regression $G\_k(\gamma) = B\_k\star$, damped conditional covariance $S\_k(\gamma) = S\_k\star$ — for every $\gamma \in (0,1]$, and the implied joint covariance is inflated

sequentially: block k's cross-block coupling scales by $1/\gamma$ through the (already inflated) conditioning covariance $\Sigma_{\{<k,<k\}}$. For **two vector blocks** the PSD condition is exactly $\gamma^2 (1 - \rho^2_{max}) > (1 - \gamma) \rho^2_{max}$, where ρ^2_{max} **is the largest squared canonical correlation between the blocks** — the scalar ρ^2 generalized to the dominant coupling mode. For K blocks the inflation, and hence the PSD threshold, **compound along the chain**.

The verification confirms each clause: the recovered $(G_k(\gamma), S_k(\gamma))$ equal the truth at every γ; the local search finds nothing above the closed form (max-gap 0); and for a random vector two-block problem with $\rho^2_{max}=0.115$ the predicted threshold $\gamma_{min}=0.302$ matches the exact PSD boundary (SPD at $\gamma=0.32$, not at $\gamma=0.28$). Three consequences, now established beyond the scalar case:

1. **Damping is undone by rescaling.** ℓ_γ "knows" the true conditional distribution at any γ; it simply expresses it through a coefficient scaled by $1/\gamma$. So *fitting* Σ to maximize ℓ_γ overcompensates — the implied joint inflates and, for strong enough coupling relative to γ, ceases to be a covariance at all. ℓ_γ **is degenerate as a free estimation objective for** $\gamma<1$; this is exactly why the sane uses are to *score* a covariance or to *transform* one (shrink the coupling, §3b) — operations that hold Σ fixed rather than letting it absorb the damping.

2. ℓ_γ **is a bias–variance dial on the scoring rule.** Because the expected score is maximized at the *inflated* $\Sigma(\gamma)$, not at $\Sigma\star$, ℓ_γ is for $\gamma<1$ **not strictly proper** for Σ: it carries a (mild, for γ near 1) bias toward inflation. Its advantage is variance: the full likelihood ($\gamma=1$) is properly centered but, through Σ^{-1} and $\log\det$, has exploding variance in high dimensions; damping trades a little properness for a large reduction in score variance. The empirically optimal evaluation γ (§5) is the one whose bias^2+variance for *ranking* is smallest.

3. **The usable range shrinks with coupling.** $\gamma_{min}(\rho^2)$ increases from 0 (weak coupling, any γ safe) to 1 (strong coupling, only the full likelihood stays PSD). This is the same near-singular Schur-complement regime in which linear damping destabilizes and geodesic damping does not (§9) — two faces of one fact: strong coupling is where naïve interpolation of the conditioning is most dangerous.

A closed-form γ^*: the coupling reliability. The same tractable model yields the *optimal* γ, not just the PSD range — for the well-behaved (predictive / estimator) use, where one damps the *estimated* coupling $\gamma\hat{G}_k$ (the operation `SchurCovariance` and Schur allocation perform), rather than free-maximizing. Choose γ to minimize the expected test predictive residual. With the OLS coupling $\hat{b}$ ($E\,\hat{b} = b$, $Var\,\hat{b} = s / (a(n-2))$), the residual $R(\gamma) = [b^2(1-\gamma)^2 + \gamma^2\,Var\,\hat{b}]\,a + s$ is minimized at

$$\gamma^* = b^2 / (b^2 + Var\,\hat{b}) = (n-2)\rho^2 / [(n-2)\rho^2 + (1 - \rho^2)], \quad \text{with} \quad \rho^2 = b^2a / (b^2a + s),$$

a **Wiener / James–Stein shrinkage of the coupling**: γ^* is its *reliability*. It rises with both the sample size and the coupling strength — $\gamma^*\to1$ (trust it → full likelihood) as $n\to\infty$ or $\rho^2\to1$, $\gamma^*\to0$ (ignore it → block-diagonal) as $\rho^2\to0$, and lies in the interior exactly when the coupling is *partially reliable*. This is the paper's thesis in closed form, verified against simulation (`research/schur_gamma_star.py`: e.g. $n=10$, $\rho^2=0.3 \to \gamma^*=0.77$, matching the simulated argmin). It is the optimal γ for the *estimator*; the high-dimensional *discrimination* γ^* (§5) is its harder, nonlinear cousin.

5. The discrimination efficiency, exactly, and the Godambe boundary

§4(2) called γ a bias–variance dial on the scoring rule. This makes that statement exact, and locates — honestly — where the high-dimensional advantage actually comes from.

The γ-Schur operator. The score is a quadratic form in the data, $\ell_\gamma(\Sigma; x) = -\frac{1}{2}[\,Ld_\gamma(\Sigma) + x^T W_\gamma(\Sigma)\,x\,] + const$, where $W_\gamma(\Sigma) = \Sigma_k T_k^T S_k(\gamma)^{-1} T_k$ assembles the damped block conditionals ($T_k = [-\gamma\hat{G}_k,\ I,\ 0]$ picks block k minus its γ-damped regression on earlier blocks) and $Ld_\gamma = \Sigma_k \log\det S_k(\gamma)$. At $\gamma=1$, $W_1(\Sigma) = \Sigma^{-1}$ *exactly* (the block-LDL^T reconstruction of the precision; verified to 1e-15). So the family deforms the precision continuously from Σ^{-1} ($\gamma=1$) to $\text{block-diag}(\Sigma_{kk}^{-1})$ ($\gamma=0$).

Discrimination SNR (exact, Gaussian). For two estimates $A,\ B$ and truth Σ_t, the per-test-sample score gap $d(x) = \ell_\gamma(A;x) - \ell_\gamma(B;x)$ under $x \sim N(0,\ \Sigma_t)$ has, by the Gaussian quadratic-form moments, $\mu_\gamma =$

$-½[\Delta Ld + tr(\Delta W\ \Sigma\_t)]$ and $\sigma^2\_\gamma = ½\ tr((\Delta W\ \Sigma\_t)^2)$, with $\Delta W = W\_\gamma(A) - W\_\gamma(B)$. Hence n_test samples rank A above B with probability ≈ $\Phi(\sqrt{n\_test} \cdot \mu\_\gamma/\sigma\_\gamma)$: the per-sample SNR μ_γ/σ_γ *is* the discrimination efficiency (closed form matches Monte Carlo to three figures).

The honest twist — under test noise alone, $\gamma=1$ **is optimal.** This SNR holds the estimates fixed and averages over test draws. In that model $\gamma=1$ maximizes the SNR in every regime we checked (well- and ill-conditioned candidates alike) — Neyman–Pearson — and damping only adds the §4 bias. **So the high-dimensional interior optimum is *not* a test-sample-noise effect.**

Where it does come from — estimate instability × shared unidentifiable structure. The score depends on the estimate through $W\_\gamma(\Sigma)$. At $\gamma=1$, $W\_1 = \Sigma^{-1}$ has sensitivity $\partial\Sigma^{-1} \sim \Sigma^{-1} \otimes \Sigma^{-1} \sim 1/\lambda\_min^2$: when the estimates carry a shared, unidentifiable, ill-conditioned component (e.g. spurious noisy cross-block coupling), a fresh data draw moves W_1 wildly and the score's verdict flips — power collapses to chance. Damping *bounds* that sensitivity (W_γ inverts only conditioning blocks). In the regime where the discriminating signal lives in the within-block correlations and the cross-block coupling is shared noise (research/metric_power.py, research/schur_godambe.py), an interior γ strictly beats both endpoints — power **0.84** (crude cross-damping) / **0.71** (the principled W_γ) at the optimum versus ≈**0.45** at $\gamma\in\{0,1\}$. The bias–variance dial of §4(2) is thus over the *estimate*, not the test sample.

The variance half of this is now analytic. To first order the training-draw variance of the score gap is $Var(D) \approx (2/n)\ tr((\nabla\_S\ D \cdot \Sigma\_t)^2)$, where $\nabla\_S\ D$ inherits the sensitivity of W_γ to the estimate. At $\gamma=1$ that sensitivity is $\partial\Sigma^{-1} \sim \Sigma^{-1} \otimes \Sigma^{-1}$, so $Var(D)$ grows like $1/\lambda\_min^4$: numerically (research/schur_spiked.py) it runs $0 \to 77$ as the truth's condition number worsens $24 \to 400$, while $\gamma=0.3$ stays bounded ($0 \to 5$). Damping caps the variance — the precise cure for the collapse. The *signal* half resists this linearization (it is second-order in the sampling noise — the shrinkage benefit), so γ^* itself stays a nonlinear quantity, characterized numerically above rather than in closed form.

The Godambe boundary. Whether ℓ_γ is even a valid estimating equation settles its status. The γ-Schur score $U\_\gamma = \nabla\_\Sigma\ \ell\_\gamma$ is unbiased at the truth — $E\_t[U\_\gamma(\Sigma\_t)] = 0$ — *only at the endpoints*: $\gamma=1$ (the full Fisher score) and $\gamma=0$ (the correctly specified block marginals, for their identified parameters). In the interior the expected score gradient is nonzero (verified: $\|\nabla\ E\ \ell\_\gamma(\Sigma\_t)\| = 0$ at $\gamma\in\{0,1\}$ and grows through the interior), because the damped conditionals are misspecified — the inflation of §4. Standard (unbiased-estimating-function) Godambe theory thus applies only at the endpoints: $\gamma=0$ carries the classical composite-likelihood efficiency loss $G\_0 = H\_0\ J\_0^{-1}\ H\_0 \preceq I\_full$ (Varin–Reid–Firth 2011); $\gamma=1$ is Fisher-efficient. The interior is neither an efficient estimator nor an unbiased score — it is a *tempering*, justified by the discrimination SNR above. This is the rigorous form of "ℓ_γ is a scoring/transform device, not a free estimation objective."

6. Schur damping is a structured shrinkage (and is complementary to spectral shrinkage)

The transform use (§3b) shrinks each conditional covariance from its full Schur complement $\hat{S}\_k$ toward the unconditional block Σ_kk with intensity $1-\gamma$. Since $\hat{S}\_k \preceq \Sigma\_kk$, this *raises* the conditional covariance, lifting the small eigenvalues of every block-sized inverse ℓ_γ performs — a James–Stein-style shrinkage applied *blockwise to the conditioning*. It is therefore **orthogonal and complementary to spectral shrinkage**: Ledoit–Wolf and the rotationally-invariant estimators (Bun, Bouchaud & Potters 2017) regularize the *eigenvalues* (basis-free); Schur damping regularizes the *cross-block coupling* (basis-dependent). Both improve conditioning, along different geometry, and can be composed: γ is to the block-conditional factorization what shrinkage intensity is to the spectrum.

7. The unifying role: one knob across many constructions

The same ℓ_γ places a scattered set of objects on **two axes of the same block factorization**:

- **Axis A — conditioning structure (sparsity):** for each block, *which / how many* earlier blocks it conditions on. Full (dense) ↔ none (block-diagonal). The axis Vecchia, GMRF, banding/tapering and nested dissection regularize along.

- **Axis B — coupling strength γ:** *how strongly* the chosen conditioning is trusted (§3). Full ($\gamma=1$) ↔ none ($\gamma=0$). The axis introduced here.

```
                     γ = 1 (full trust)       γ ∈ (0,1)              γ = 0 (no trust)
full conditioning   exact Gaussian likelihood ┌────────────────────┐  block composite
(dense)             = MVO  w ∝ Σ^{-1}1        │  SCHUR PSEUDO-LIK  │   likelihood
                                              │    (this work)     │   = HRP-style allocation
sparse conditioning block-Vecchia / GMRF      │  Schur + sparsity  │        (block-marginal,
(Vecchia/GMRF)      likelihood (exact conds.) └────────────────────┘         same column)
nested dissection   = Schur-complement elimination of the (sparse) blocks
```

- **Vecchia / GMRF / nested dissection** (Vecchia 1988; Katzfuss–Guinness 2021; Pan et al. 2024; Rue–Held 2005; George 1973) use the *exact* conditioning ($\gamma=1$) and regularize by **sparsity**. The Schur pseudo-likelihood adds the orthogonal **strength** axis; the two compose.
- **Composite / pseudo-likelihood** (Besag 1975; Lindsay 1988; Varin–Reid–Firth 2011) is the $\gamma=0$ column; ℓ_γ is its one-parameter deformation back toward the joint likelihood.
- **Portfolio allocation, the same algebra transported:** minimum-variance ($w \propto \Sigma^{-1}1$) is the $\gamma=1$ full-conditioning corner, hierarchical risk parity (López de Prado 2015/2016) the $\gamma=0$ corner, and **Schur complementary allocation** (Cotton 2024) damps *the same Schur complement with the same* γ between them. The Schur pseudo-likelihood is its evaluation/estimation mirror — identity, not analogy; Antonov–Lipton–de Prado (2024) on HRP stability explains the interior dominance.

One sentence: γ **is one knob that, at its endpoints, recovers the joint likelihood, composite likelihood, block-Vecchia/GMRF likelihoods, MVO and HRP — and in its interior is the common bridge all of these lack.**

8. Broad applicability

The construction needs only three ingredients — a Gaussian (or Gaussian-pseudo) likelihood, a block partition, and the conditional/Schur factorization of §2 — so the coupling-strength knob is **generic to Gaussian-block models**, not specific to covariance estimation or finance:

- **Spatial statistics / Gaussian processes.** Vecchia and GMRF likelihoods already condition each location on a neighborhood; γ adds a robustness dial on *how strongly* those neighbor relations are trusted — orthogonal to, and composable with, the existing choice of neighborhood size/ordering.
- **Gaussian graphical models / precision estimation.** With blocks as cliques or neighborhoods, γ damps the conditional-dependence strength, a continuous relaxation between the full penalized likelihood and node-/block-wise pseudo-likelihood (Besag; Meinshausen–Bühlmann).
- **Longitudinal / panel / state-space models.** With blocks as time slices, γ interpolates between treating slices independently ($\gamma=0$) and the full temporal joint ($\gamma=1$) — a tempering of the temporal coupling.
- **General composite likelihood.** Anywhere composite likelihood is used for tractability (Varin– Reid–Firth's survey spans genetics, networks, multivariate survival, image analysis), γ is a generic dial from the composite ($\gamma=0$) toward the full ($\gamma=1$) likelihood, with §4's caveat that it is to be used for scoring/tempering, not free estimation.
- **High-dimensional model scoring.** As a *score*, ℓ_γ is a high-dimension-robust, nearly-proper alternative to the log score for any of the above — the use we test directly here.

What we prove is the two-block characterization (§4) and demonstrate the covariance-evaluation case (§9); the wider applicability is a claim about the *mechanism* (damping a conditional factorization shared by all these models), flagged as such, not a measured result in each domain.

9. Geometry, and results

Linear vs geodesic damping. Damping interpolates each conditional covariance between Σ_{kk} and S_k. Linear damping is $S_k(\gamma) = (1-\gamma)\Sigma_{kk} + \gamma S_k$; **geodesic** damping moves along the affine-invariant SPD geodesic $\Sigma_{BD} \#_\gamma \Sigma = \Sigma_{BD}^{1/2}(\Sigma_{BD}^{-1/2}\, \Sigma\, \Sigma_{BD}^{-1/2})^{\gamma}\, \Sigma_{BD}^{1/2}$ (the Fisher–Rao geodesic, reusing the `GeodesicEwaCovariance` step), lifting small eigenvalues *multiplicatively*. They agree to first order and at the endpoints but differ near a singular Schur complement.

Results (reproducible; rankings are ensemble-dependent — see `papers/evaluation_and_generation_review.md`):

- γ **traces a regime-dependent optimum** (`research/metric_power.py`). Power climbs to $\gamma=1$ when the coupling is reliable (Neyman–Pearson); falls toward $\gamma=0$ in the high-dimensional noisy regime; and at *partial* reliability an **interior** γ **strictly dominates both endpoints** (≈0.84 vs ≈0.43) — the bias–variance optimum of §4(2), and the likelihood analogue of Schur allocation beating both HRP and MVO.
- **Geodesic stability.** Near a singular Schur complement (strong coupling — the $\gamma < \gamma_{min}$ danger zone of §4(3)) linear damping dips (≈0.64) while geodesic damping holds full power.
- **The motivating failure, economically** (Ken French 100 portfolios, rolling minimum-variance OOS volatility, $p=100$, $n=60$; `research/oos_equity.py`). Not a test of ℓ_γ as an equity estimator — we are explicit about that — but a confirmation of the catastrophe it avoids: inverting/unregularized estimators blow up (annualized vol 9.6 empirical, 6.8 Huber, 1.3×10^6 Tyler) while well-conditioned shrinkage stays at 0.141 (vs 0.129 oracle, 0.228 equal-weight). In high dimensions the reward is for *conditioning* — for not trusting the raw inverse, which is what $\gamma<1$ does.

10. Limitations and open problems

- **Closed-form γ^*.** For the *estimator/predictive* use this is now solved in the tractable two-block model (§4): $\gamma^* = (n-2)\rho^2/[(n-2)\rho^2 + (1-\rho^2)]$, the coupling reliability, verified against simulation. For the high-dimensional *discrimination/scoring* use the picture is partial: §5 gives the exact test-sample SNR ($\gamma=1$ optimal), the Godambe boundary (unbiased only at $\gamma\in\{0,1\}$), and the first-order training-draw *variance* (verified $1/\lambda_{min}^4$ blow-up at $\gamma=1$, bounded for $\gamma<1$). The open gap is the discrimination *signal* term — second-order in the sampling noise (the shrinkage benefit) — so a closed-form discrimination γ^* needs beyond-first-order (exact Wishart-moment / large-deviation) analysis; it is so far characterized only numerically (the interior optimum of §5).
- **Compounded threshold.** §4 has the two-block ρ^2_{max} PSD form and numerical evidence that the K-block threshold compounds along the chain, but not yet a closed-form chain formula.
- **Optimal γ.** The evaluation/tempering optimum $\gamma^*(p,\ n,\ \text{spectrum},\ \text{blocks})$ is characterized only empirically; a plug-in rule from the effective rank / coupling ρ^2 is open. (We are wary of a *trained* selector after the sibling recommender failed to generalize across novel generative families — `research/oos.py`.)
- **Block ordering and choice**, and composition with the Axis-A sparsity choice (the "Schur + sparsity" cell), remain to be studied; like Vecchia, ℓ_γ depends on the partition/ordering.
- **Scaling.** Recursive block inversion gives $O(p\cdot b^2)$; a Russell-scale ($p\approx2000$) real-data study, where the separation should be largest, is set up to run.
- **Falsification.** The thesis predicts the interior-γ dominance should *vanish* when the coupling is fully reliable (LKJ, large η, low dimension → $\gamma^*\to1$) or pure noise (hard Marčenko–Pastur bulk → $\gamma^*\to0$); an interior optimum in a known-fully-reliable regime would contradict it.

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References

López de Prado (2015/2016); Antonov, Lipton & López de Prado (2024, SSRN 4748151); Cotton (2024, arXiv:2411.05807); Besag (1975); Lindsay (1988); Varin, Reid & Firth (2011); Vecchia (1988); Katzfuss & Guinness (2021); Pan, Abdulah, Genton & Sun et al. (2024); Rue & Held (2005); George (1973); Meinshausen & Bühlmann (2006); Ledoit & Wolf (2004, 2012); Bun, Bouchaud & Potters (2017); James & Stein (1961); Gneiting & Raftery (2007); Scheuerer & Hamill (2015). Full annotated lists in `papers/evaluation_and_generation_review.md` and `papers/covariance_evaluation.md`.